Examining 3D Self-Supervised Methods for Medical Imaging

Jonathan Brown (jcb472) [1]

[1]Cornell University

Motivation

Problem:

- Medical imaging suffers from limited labeled data
- Expert annotation is **expensive and time-consuming**

Paper Insight:

- *"3D Self-Supervised Methods for Medical Imaging"* proposes learning from **unlabeled 3D scans**
- Uses **self-supervised proxy tasks** to pretrain models

Our Goal:

- Re-implement key self-supervised methods:
  - Rotation Prediction
  - Relative Patch Location (RPL)
  - Jigsaw Puzzles
  - Contrastive Predictive Coding (CPC)
  - Exemplar Networks
- Evaluate whether similar **data efficiency improvements** can be reproduced

Hypothesis:

- Self-supervised pretraining improves performance in **low-data regimes**

Methodology

Outline:

- Pretrain a shared **3D encoder** using self-supervised tasks
- Transfer encoder into a **3D U-Net** architecture
- Fine-tune on a downstream task using varying labeled data fractions

Real world Task:

- **Hippocampus segmentation** from 3D MRI scans
- Predict voxel-level labels to identify anatomical structure

Evaluation:

- Compare pretrained models vs **training from scratch**
- Measure performance using **Dice score**
- Analyze **data efficiency** across labeled fractions

Constraints:

- Large dataset size → limited storage and compute
- Used **subset of data** instead of full dataset
- Introduced higher variance in results

Mitigation:

- Used **multiple random seeds**
- Averaged results to improve reliability

Dataset Visualization

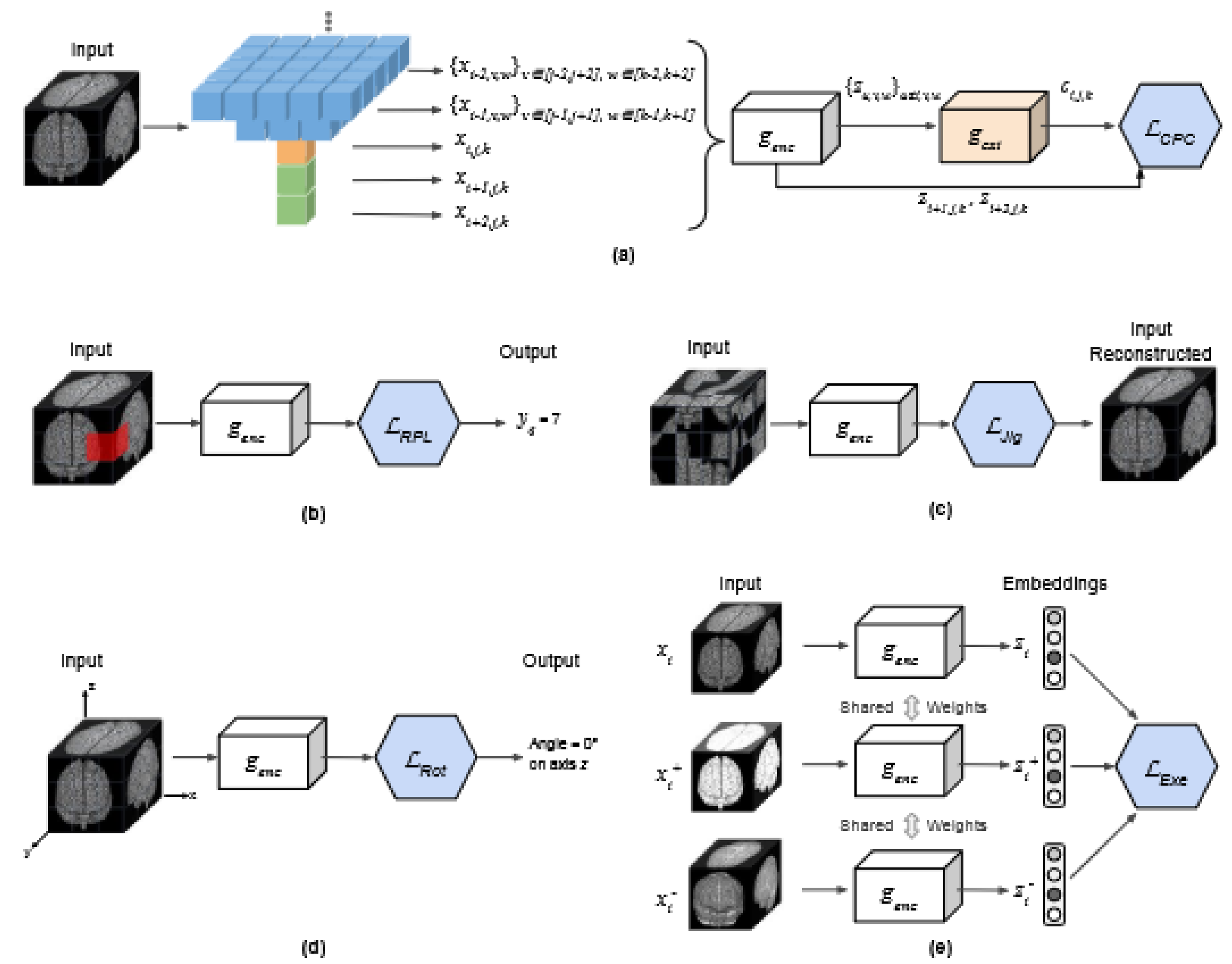
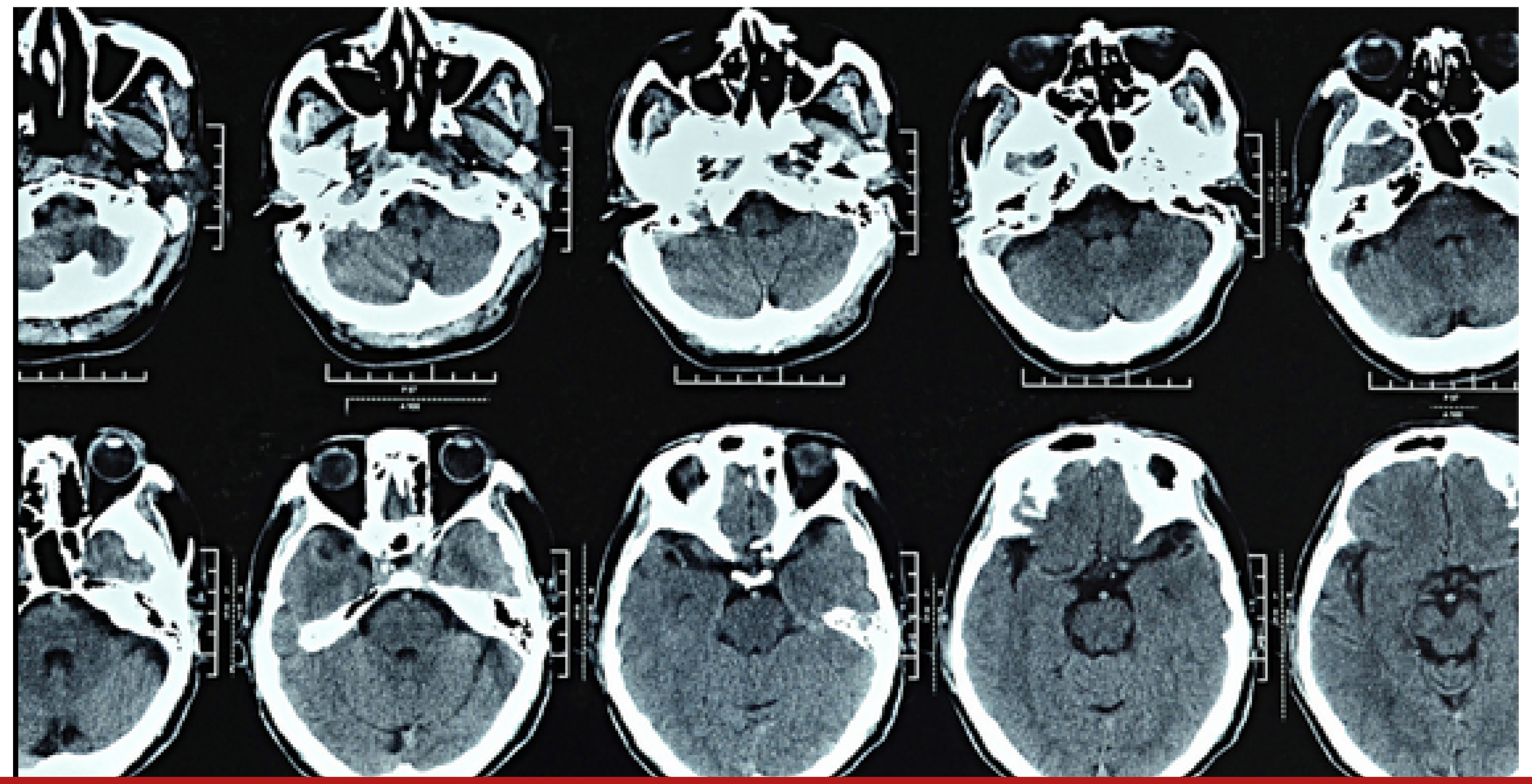


Figure 1: (a) 3D-CPC: each input image is split into 3D patches, and the latent representations $z_{i+1,j,k}, z_{i+2,j,k}$ of next patches $x_{i+1,j,k}, x_{i+2,j,k}$ (shown in green) are predicted using the context vector $c_{i,j,k}$. The considered context is the current patch $x_{i,j,k}$ (shown in orange), plus the above patches that form an inverted pyramid (shown in blue). (b) 3D-RPL: assuming a 3D grid of 27 patches ($3 \times 3 \times 3$), the model is trained to predict the location y_q of the query patch x_q (shown in red), relative to the central patch x_c (whose location is 13). (c) 3D-Jig: by predicting the permutation applied to the 3D image when creating a $3 \times 3 \times 3$ puzzle, we are able to reconstruct the scrambled input. (d) 3D-Rot: the network is trained to predict the rotation degree (out of the 10 possible degrees) applied on input scans. (e) 3D-Exe: the network is trained with a triplet loss, which drives positive samples closer in the embedding space (x_i^+ to x_i), and the negative samples (x_i^-) farther apart.

Figure 1. How the self-supervised methods work on the 3D data



Results

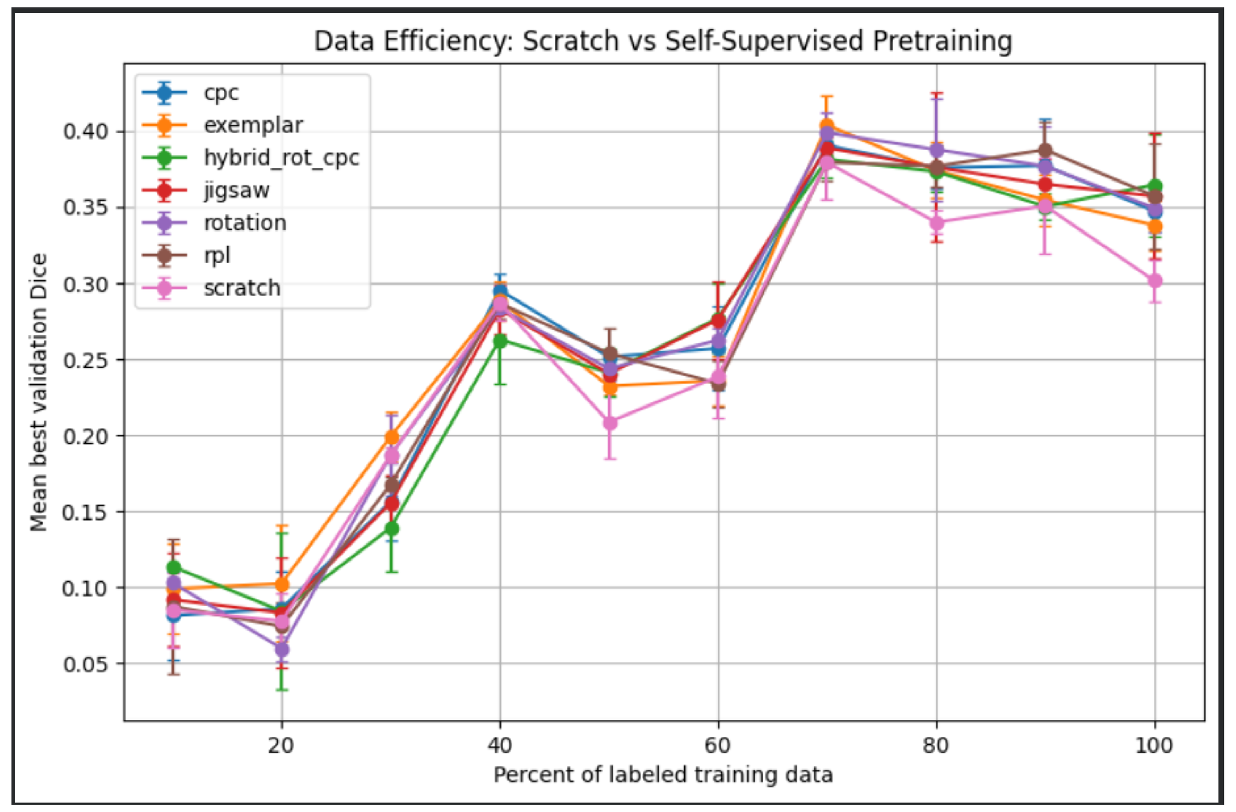


Figure 3. Data efficiency comparison of self-supervised methods vs training from scratch across labeled data fractions.

Key Observations:

- Performance improves as the percentage of labeled data increases
- All methods converge at high data fractions (80–100%)
- Self-supervised methods are competitive with training from scratch
- Expected: Strong advantage of self-supervised learning in low-data regimes
- Observed: Similar performance across methods with limited separation
- Reduced dataset size increases variance
- Limited compute and batch sizes affect contrastive methods (e.g., CPC)
- Results still follow overall trend of performance saturation

Future Work

- Run longer pretraining with a larger unlabeled dataset to better test whether self-supervised learning improves data efficiency.
- Compare Rotation-only, CPC-only, and hybrid Rotation + CPC pretraining directly to see whether the hybrid model actually improves transfer.
- Tune the hybrid loss more carefully so that one objective does not dominate the other during pretraining.
- Run experiments across more random seeds and labeled-data fractions to reduce variance in the results.
- Add qualitative mask visualizations to compare predicted hippocampus segmentations against ground-truth masks.

References

- Taleb et al *3D Self-Supervised Methods for Medical Imaging*. Advances in Neural Information Processing Systems, 2020.
- Ronneberger, O. et al *U-Net: Convolutional Networks for Biomedical Image Segmentation*. MICCAI, 2015.
- Medical Segmentation Decathlon. *Medical Image Segmentation Benchmark Datasets*.
- PyTorch. *An Imperative Style, High-Performance Deep Learning Library*.

jcb472@cornell.edu